

Name: Elijah Henderson, Odion Odion-Davis, Kordell Armstrong **Date:** 2/3/2026 **Class:** 7

MyDesign Portfolio

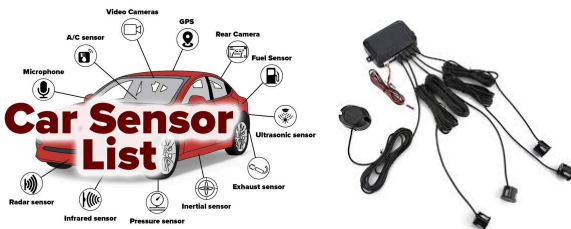
Element B Initial Template

How to use this template

Element B: Documentation and Analysis of Prior Design Attempts

Prior Design Attempt #1 - <Car Sensors #1>

<https://www.autoracing1.com/pl/468094/defective-sensors-leading-to-car-crashes-how-do-car-accident-lawyers-tackle-such-cases/>



This design attempt was made for many sensors that are used in cars. These sensors are normally made to keep cars from colliding with other cars. If you have an electric car you may see that when you're driving and your car is either getting too close to another car it may start to make a sound implying that you're getting too close to that car. As in the image we can see there are plenty of sensors that a car needs to make sure that you're safe on the road.

Although this was helpful it still didn't fix the problem because people still got into crashes and people said sometimes the sensor would malfunction so they wouldn't even know there are problems with the car like less gas or when someone in a car was speeding towards them. Our sensor for the drink machine can't malfunction because people would get upset and start saying the machine is completely broken.

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Prior Design Attempt #2 - <Motion Detectors #2>

<https://northeastsecuritysolutions.com/creative-alarm-design-ideas/>



This is a motion detector that is used to see when movement is going on in your house. If in the scenario where someone breaks into your house, you're asleep, and you didn't turn the alarm on, this can detect the motion so you can be alerted to call the police in case someone is trying to rob you. Some of these do alert the police so they can come to your door as fast as possible.

This design attempt is bad at times and I say this because if you are asleep sometimes it doesn't alert the police or the person in the house meaning someone can easily come into your house and still things if you don't have any way to wake you in a situation like that. If the motion detector made noise when things like this happened it could keep everyone in the house safe and alert the police to come help. For our design I don't think we need it to make noise but definitely a noticeable signal that informs the person that the syrup is low so they can ask someone to put more syrup in the machine.

Prior Design Attempt #3 - <insert name of prior design attempt #3>

<https://www.nature.com/articles/s41598-025-03056-x#:~:text=Our%20approach%20utilizes%20fundamental%20building,designs%20reported%20in%20the%20literature>



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The ring camera is made to put on your front door so when people come to your door you are alerted and you can make sure who it is. The ring camera is extremely successful for everyone who has them now because it truly works very well.

The device was very well made and could be a good way for us to make a sensor for the drink machine. The ring sensor has a way of alerting people and a good signal which is important for our design to go well.

Prior Design Attempt #4 - < Sensory Technology in Cameras >



House cameras require sensory information to know when someone walks in, out, or around the house. The reason is for the safety of the house and its members. In case an intruder tries to break inside of the house during the night and the alarm can go off, but it has its downside such as intruders can easily wear masks or dress in all black, hiding their faces.

An example of the design is the automatic garage door opener which was designed to detect rigid objects. The technology was a mechanism that used pressure when the door hit an obstruction(here is a source that shows that effect of this design)

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Prior Design Attempt #5 - < Fire Alarm #5>

<https://www.acfalarm.net/blog/2023/12/15/step-by-step-guide-how-fire-alarm-systems-detect-and-respond-to-fires>



Fire alarms alert you if there is a fire or if there is heavy smoke in the area to alert you to leave where it's happening.

If there is a fire and the alarm doesn't go off, people may not realize their danger until it's too late. Smoke and fire can spread quickly, increasing the risk of injury. This can delay emergency response and make the fire harder to control. And people can not get out in time and people's lives can be in danger. If there is a lot of smoke that can get in people's lungs and make it dangerous for them to breathe and if the fire alarm doesn't go off nobody will know.